

Malware Analysis Report For elf-Linux-hppa-bash

Binary Type

ELF (Executable and Linkable Format). (Approved)

Analyzing File

- **File Name:** elf-Linux-hppa-bash
- **File Extension:**
- **File Type:** ELF (Executable and Linkable Format).
- **MD5 Hash:** 6031df417ad2e8e279bab331597ad32b
- **SHA256 Hash:** a50a8b9f592606816df0f4139d5f56a400f62535ed48085757b3a1d1925d211e

File Identification

File detected as a valid Linux executable.

ELF File Analysis

ELF File Analysis: True

ELF File Imports

- libcurses.so.5
- libdl.so.2
- libc.so.6

ELF File Exports

No Exported functions were found.

String Analysis

ASCII Strings

- /lib/ld.so.1
-)l*5dyv,
- uus F|
- X0PCs0
- ac.'6Q
- H\jh"
- 'Q2]*;'
- PuMc7d~
- *_LqV;u
- S+,CjPF
- \$Ri;G>GM\$
-)l\$o>>M

- q:=|v~
- F^(`AO"
- libncurses.so.5
- GLOBALOFFSETTABLE
- JvRegisterClasses
- tgetflag
- tgetent
- **gmon_start**
- tgetnum
- tgetstr
- libdl.so.2
- dlopen
- dlclose
- dlerror
- libc.so.6
- setuid
- socket
- fflush
- strcpy
- setgrent
- pathconf
- /Oputc
- setlocale
- mbrtowc
- tolower
- gai_strerror
- strsignal
- strncmp
- getpwent
- strrchr
- setservent
- regexec
- dcgettext
- wctomb
- getpwuid
- closedir
- strncpy
- sigprocmask
- regfree
- getgrent
- unlink
- putchar
- select
- towupper
- strpbrk
- getpid
- strftime
- iswalnum
- iswlower
- strtod
- _ctypegetmbcur_max
- strtol

- endpwent
- isatty
- getpwnam
- getppid
- wcschr
- iswctype
- __mbrlen
- sigemptyset
- getaddrinfo
- memset
- strstr
- _ermolocation
- tcsetattr
- confstr
- memcmp
- endgrent
- mbtowc
- ttyname
- clearerr
- unsetenv
- __fxstat64
- vsnprintf
- sigaddset
- iswupper
- putenv
- __sigsetjmp
- stdout
- getrusage
- _strtol/internal
- strlen
- memcpy
- fclose
- tcsetpgrp
- wcscmp
- strtoul
- siglongjmp
- killpg
- strcat
- strcasecmp
- setpgid
- getpeername
- mbsinit
- getgid
- asprintf
- __lxstat64
- opendir
- __xstat64
- _ctypeb_loc
- regcomp
- wcwidth
- stderr
- sigdelset

- setrlimit64
- group_member
- getgroups
- getuid
- readlink
- getegid
- strncasecmp
- fileno
- gethostname
- getcwd
- fwrite
- gettimeofday
- sigaction
- geteuid
- waitpid
- localtime
- lseek64
- fdopen
- readdir64
- tcgetattr
- _ctypetoupper_loc
- _ctypetolower_loc
- freeaddrinfo
- setvbuf
- wcscoll
- getrlimit64
- wcsdup
- getdtablesize
- memmove
- fopen64
- endservent
- wcsrtombs
- bindtextdomain
- setgid
- _strtoullinternal
- tcgetpgrp
- _libcstart_main
- setpwent
- ferror
- wcslen
- strcoll
- vfprintf
- wcsncmp
- sysconf
- getservent
- __environ
- _edata
- _bssstart
- initializejobcontrol
- sys_error
- maybe_makeexport_env
- emacsmetakeymap

- eval_doc
- dispose_word
- shdoublequote
- rclear_screen
- rcompletionquery_items
- rterm_IC
- lastcommandexit_signal
- rinsert_typein
- times_builtin
- rcopytext
- lastcommandsubst_pid
- ifs_firstc
- rfindcompletionword
- gethistoryevent
- rlviyank_to
- promptstringpointer
- rdisablettysignals
- getjobby_pid
- listminuso_opts
- makeselectcommand
- emacsstandardkeymap
- disableprivmode
- rttystatus
- rtildeexpand
- builtin_doc
- rlnsearch_callback
- currenthostname
- sh_canonpath
- putcommandnameintoenv
- discardunwindframe
- runningunderemacs
- dequote_string
- posixreadlineinitialize
- stringlistdollar_star
- executecommandinternal
- printf_builtin
- rinitlinestate
- rcopyforward_word
- shcontainsshell metas
- sh_exit
- namedfunctionstring
- it_groups
- reader_loop
- sv_ignoreeof
- tt_setonechar
- readlineinternalchar
- no_args
- updateexportenv_inplace
- reraseentireline
- progcompletionenabled
- sccs_version

- readlinetoplevel
- extractarrayassignment_list
- rbackwardword
- print_command
- local_builtin
- progcomp_size
- rltransposechars
- thecurrentworking_directory
- ansiexpand
- rlkeyseqcxalloc
- rkillword
- indirectionlevel/string
- array_dispose
- rltyunsetdefaultbindings
- setsignalhandler
- makesubshellcommand
- arithfordoc
- lastcommandexit_value
- sv_histignore
- rlcompletionmode
- killcurrentpipeline
- history_base
- dstack
- politeformat
- xdupmbstowcs
- makebuiltinargv
- rlvisstart_inserting
- bashaddhistory
- enable_doc
- logout_doc
- rlinput_queued
- rlparsingconditionalizedout
- strlisttoword_list
- rloverwritemode
- remember_args
- disposecondnode
- rlstrip_prompt
- rlfilenameword_characters
- while_doc
- newvarcontext
- setbashinput_fd
- substring
- bashtildefind_word
- rbackwardchar_search
- lastshellbuiltin
- history_length
- _libccsu_fini
- rl_initialize
- list_opttype
- numberofargs
- prog_completes

- *arraydisposeelement*
- *rlforwardchar*
- *make_word*
- *setcurrentprompt_level*
- *rlviappend_mode*
- *listrestof_args*
- *source_builtin*
- *rlcompletiontype*
- *builtin_error*
- *rlgetscreenize*
- *bindvariablevalue*
- *pushd_builtin*
- *xpg_echo*
- *remove_history*
- *rlenablemetakey*
- *rlconvertmetacharstoascii*
- *rlviredo*
- *tt_setnocanon*
- *getvariablevalue*
- *rlterm_im*
- *rlgnoreadline_p*
- *freebufferedstream*
- *rlfunmapnames*
- *shopt_setopt*
- *sh_readonly*
- *rlsetmarkat_pos*
- *it_signals*
- *rlsavestring*
- *shell_execve*
- *wlcache*
- *sv_histchars*
- *continue_doc*
- *pat_subst*
- *braceargseparator*
- *start_pipeline*
- *rl_dispatching*
- *rlresetterminal*
- *hash_walk*
- *justonecommand*
- *rladdfunmap_entry*
- *rldirectoryrewrite_hook*
- *find_function*
- *rlcompletemarkdirectories*
- *commandstringindex*
- *getdirstackfrom_string*
- *write_history*
- *rldisplayprompt*
- *read_octal*
- *rlvitilde_expand*
- *rlsetkeyboardinputtimeout*
- *rlunbindcommandinmap*

- *rlout_stream*
- *rl_instream*
- *xstrchr*
- *rldisplayfixed*
- *allvisiblevariables*
- *rl_forward*
- *replacehistoryentry*
- *rlshowchar*
- *rlresizeterminal*
- *rlbindkeyifunboundinmap*
- *rlresetsscreen_size*
- *sh_getopt*
- *rl_message*
- *replacehistorydata*
- *umask_doc*
- *rlmacrodumper*
- *array_rshift*
- *rlvinitializeline*
- *findvariableinternal*
- *ifsfirstcilen*
- *signal_name*
- *find_variable*
- *internal_strmatch*
- *rl_abort*
- *no_options*
- *initialize_traps*
- *parseandexecute*
- *sh_notfound*
- *endjobcontrol*
- *unbind_args*
- *rlinitterminalio*
- *umask_builtin*
- *rlonnewlinewith_prompt*
- *hash_string*
- *sh_notbuiltin*
- *quoteglobbingchars*
- *dist_version*
- *executing*
- *subshell_argv*
- *ttsetattr*
- *sourcesearchescwd*
- *rlnonincreverse_search*
- *getcurrentuser_info*
- *rlexecutenext*
- *rlcopyundoentry*
- *rlmacrobind*
- *rlcontrol_keypad*
- *rlgetkeymap_name*
- *command_connect*
- *rlkscxt*
- *it_shopts*

- `select_doc`
- `rlfreesavedhistory_line`
- `rltermforwardchar`
- `popd_builtin`
- `rlbyteoriented`
- `shownameattributes`
- `historywritetimestamps`
- `break_doc`
- `bashreadlineinitialized`
- `rlhorizontalscrollmode`
- `withinputfrom_stdin`
- `makewordfrom_token`
- `noinvisiblevars`
- `checkhashedfilenames`
- `rlvariablevalue`
- `historyexpansionchar`
- `maybeexecutefile`
- `hist/astline_added`
- `saveparserstate`
- `change_flag`
- `true_doc`
- `currentreadlineline_index`
- `describe_pid`
- `top/level/mask`
- `rlonnew_line`
- `rladjust_point`
- `it_keywords`
- `maybeappendhistory`
- `allexportedvariables`
- `set_shellopts`
- `theprintedcommand`
- `signal_names`
- `delete_job`
- `rlterm_ei`
- `maybesetsigchld_trap`
- `unquotedglobpattern_p`
- `push_stream`
- `rlbindkeyinmap`
- `lastasynchronouspid`
- `rlbasicwordbreakcharacters`
- `suspend_doc`
- `findshellbuiltin`
- `array_patsub`
- `rlpendinginput`
- `rlpage_completions`
- `builtinaddressinternal`
- `initializeterminatingsignals`
- `rlpossiblecontrolprefixes`
- `rl eof_char`
- `rlsuppress_redisplay`

- *give**terminal**to*
- *rlun**bind**function**in**map*
- *rlcopy**keymap*
- *rlreverse**search**_history*
- *alias**expand**all*
- *rl**filenamedequoting**_function*
- *parseandexecute**_level*
- *execute**_command*
- *rl**_done*
- *shouldignore**glob**_matches*
- *strlist**prefix**suffix*
- *timetocheck**_mail*
- *setbuffered**stream*
- *rlrestore**state*
- *rlinhibit**completion*
- *historyexpansion**inhibited*
- *pipeline**_pgrp*
- *rlkeysequence**_length*
- *rlset**insert**mode*
- *rememberon**history*
- *rlignore**completion**_duplicates*
- *strsub*
- *pwd**_doc*
- *enable**hostname**completion*
- *tt**_setcbreak*
- *pop**_context*
- *let**_builtin*
- *dispose**_variable*
- *ttattr*
- *rloutput**some**chars*
- *command**word**completion**_function*
- *sh**_regmatch*
- *allarray**variables*
- *bashbrace**completion*
- *read**_history*
- *makecond**node*
- *rlcompletion**append**_character*
- *eofencountered**limit*
- *posix**_initialize*
- *rlkeyseq**chain**dispose*
- *rl**_yank*
- *maxinput**history*
- *initialize**bash**input*
- *pop**var**context*
- *rluniversal**argument*
- *check**_identifier*
- *shsingle**quote*
- *list**running**jobs*
- *zsyncfd*
- *historyquotes**inhibit**_expansion*
- *strlist**from**word**_list*

- `termsig_sighandler`
- `bashdefaultcompletion`
- `binary_test`
- `ps2_prompt`
- `history_get`
- `uinttostr`
- `current_user`
- `currentcommandline_count`
- `rlarg_overflow`
- `rllyankpop`
- `returncatchflag`
- `rlenable_keypad`
- `initialize_aliases`
- `rlpossiblecompletions`
- `rlbindingkeymap`
- `exit_builtin`
- `rlrestoreprompt`
- `suspend_builtin`
- `rlignoresomecompletionsfunction`
- `rlgenericbind`
- `rlfilenamequoting_desired`
- `tt_seteightbit`
- `rlsetkeymap`
- `rlvichange_char`
- `build_version`
- `readlineinternalsetup`
- `rlgetcfunction`
- `getarrayvalue`
- `word_split`
- `file_status`
- `setsigchldhandler`
- `rlvisibleprompt_length`
- `historylinesthis_session`
- `rlterm_cleol`
- `it_arrayvars`
- `tilde_initialize`
- `exec_doc`
- `closebufferedfd`
- `allvariablesmatching_prefix`
- `rlruboutor_delete`
- `rlastvpos`
- `rltranslatekeyseq`
- `currentreadlineprompt`
- `rlsearch_dispatch`
- `it_services`
- `errortracemode`
- `findreservedword`
- `shsetlinesandcolumns`
- `xtraceprintforcommandhead`
- `duplicatebufferedstream`
- `getshoptoptions`

- *rlpushexecutingmacro*
- history_builtin
- zreadc
- rlvisearch
- variable_context
- *rlenable_meta*
- usercommandmatches
- strvec_dispose
- *rldispatch_subseq*
- historylinesin_file
- rlexecutingmacro
- *rlbackwardchar*
- *rlinput_available*
- hashing_enabled
- ulimit_doc
- currentcommandsubst_pid
- *rldeprepterm_function*
- array_remove
- *rldumpvariables*
- compgen_builtin
- *rlchar_value*
- traptosighandler
- *rlcharacterlen*
- type_builtin
- *setpipestatusarray*
- strlist_dispose
- it_jobs
- rl_rubout
- err_badarraysub
- *rladddefun*
- setlocalevar
- addhistorytime
- sh_optopt
- return_builtin
- *svstrictposix*
- pop_scope
- *rldoinganundo*
- *rlcatchsigwinch*
- historyinhibitexpansion_function
- clear_history
- cdspelling
- return_catch
- strvec_copy
- savetokenstate
- getmaxchild
- makewordlist
- *rleditingmode*
- *rlforwardsearch_history*
- makesimplecommand
- command_errstr
- historyisstifled

- *getdirectorystack*
- *rlvireseflast*
- *source_doc*
- *rlbegof_line*
- *rl_complete*
- *rlbackspace*
- *popd_doc*
- *rl_redisplay*
- *lasthistoryline*
- *unset_builtin*
- *rlclearpending_input*
- *bashservicenamecompletion_function*
- *rlinsert_char*
- *compgen_doc*
- *getmaxgroups*
- *ttocanon*
- *rlgetkeymapbyname*
- *streamonstack*
- *expandarithstring*
- *rlkillregion*
- *sh_mktmpfd*
- *unboundvarsis_error*
- *disown_doc*
- *progcomp_remove*
- *bindvarto_int*
- *export_builtin*
- *bind_builtin*
- *noclobber*
- *setttystate*
- *expandpromptstring*
- *remove_alias*
- *hist/asline_pushed*
- *parse_shellopts*
- *initializeshell/builtins*
- *rlcleartoel*
- *commandexecutionstring*
- *maybeaddhistory*
- *indirection_level*
- *rlcopybackward_word*
- *othertokenalist*
- *sh_stat*
- *cmd_init*
- *setsignalignored*
- *currentcommandnumber*
- *absolute_pathname*
- *string/istinternal*
- *rlargsign*
- *strvec_sort*
- *printfundlist*
- *resetsignalhandlers*
- *rlterminalcaninsert*

- *rterm_ip*
- *wantpendingcommand*
- *history_search*
- *sh_invalidopt*
- *waitsignalreceived*
- *restoredefaultsignal*
- *setjobcontrol*
- *unbindarrayelement*
- *array_create*
- *seterrortrap*
- *rbindkeyseqinmap*
- *shift_builtin*
- *nosymboliclinks*
- *builtin_usage*
- *readonly_builtin*
- *rlclearsignals*
- *tildeexpansionpreexpansion_hook*
- *kill_doc*
- *fgpercentdoc*
- *interrupt_immediately*
- *sh_mktmpname*
- *disposefdbitmap*
- *rlinsertcomment*
- *rl_newline*
- *dowordassignment*
- *rlinternalcharcleanup*
- *getminuso_opts*
- *setdollarvars_unchanged*
- *rterm_pc*
- *defaultttyjob_signals*
- *rlreadlinestate*
- *array_insert*
- *historysubstchar*
- *rlcompleterwordbreakcharacters*
- *unset_doc*
- *sv_histtimefmt*
- *unwindprotectmem*
- *wdcache*
- *cleanupthepipeline*
- *sh_invalidsig*
- *wait_builtin*
- *exec_builtin*
- *dirs_builtin*
- *parsestringtowordlist*
- *rlenableparenmatching*
- *copyfunctiondef_contents*
- *popdollarvars*
- *rldispatch*
- *sourcelevel*
- *getcurrentprompt_level*
- *findstringin_alist*

- describe_command
- trap_doc
- makewordflags
- *rlrubout_char*
- phash_search
- rlvisubst
- *rlcompletemarksymmlink_dirs*
- bashinputfd_changed
- *rlsettheline*
- test_doc
- allocworddesc
- *rlinsertmode*
- *rldirectorycompletion_hook*
- sv_opterr
- stupidlyhackspecial_variables
- *rlsearch_getchar*
- setifs
- stream_list
- *setitemlistdirty*
- fatal_error
- subshell_argc
- preprocessline
- printarithcommand
- *rlcompletionmatches*
- *rltabinsert*
- stringrestof_args
- printvarfunction
- loop_level
- extractarithmeticsubst
- localetrans
- arraytostring
- rl_modifying
- history_tokenize
- *rlredisplayaftersigwinch*
- find_flag
- *rlmakepromptfor_search*
- fg_builtin
- disposeexecredirects
- return_doc
- *rlgetcharlen*
- localexportedvariables
- unsetbashinput
- *rlabort_internal*
- *lOstdin_used*
- *rlcopyregiontokill*
- *rlmove_vert*
- *rlmeta_flag*
- makecasecommand
- rlundolist
- printf_doc
- conditional_doc

- *rlreset_argument*
- *rlmakebare_keymap*
- *initializeshelloptions*
- *failglobexpansion*
- *getworkingdirectory*
- *rlgetnext_history*
- *findtempenvvariable*
- *makefunctiondef*
- *rlterm_dc*
- *command_builtin*
- *shunsetnodelay_mode*
- *staticshell/builtins*
- *setvarread_only*
- *rlcompletionmarkssymlinkdirs*
- *colon_doc*
- *cd_doc*
- *rlinvokingkeyseqs*
- *function_doc*
- *historygettime*
- *ttcbreak*
- *time_doc*
- *zwrite*
- *_getenv*
- *stdin_redir*
- *get_exitstat*
- *rlreadinit_file*
- *viinsertionkeymap*
- *builtin_builtin*
- *rlfilenamecompletion_desired*
- *builtin_address*
- *history_expansion*
- *rlterm_goto*
- *restoreoriginalsignals*
- *maybesaveshell_history*
- *unsetsigwinchhandler*
- *rlin_stream*
- *rlcompletiondisplaymatcheshook*
- *resetshellflags*
- *rlendundo_group*
- *beginunwindframe*
- *disown_builtin*
- *rl_crlf*
- *setdebugtrap*
- *rlcopyundolist*
- *runreturntrap*
- *declare_doc*
- *rlrestorettysignals*
- *checkbinaryfile*
- *disposesaveddollar_vars*
- *sh_invalidoptname*
- *rlscreenchars*

- `rlnamedfunction`
- `listalljobs`
- `rlbindkeyifunbound`
- `rlunixword_rubout`
- `alias_expand`
- `makelocalvariable`
- `disallowfilenameglobbing`
- `alreadymakingchildren`
- `resetinternalgetopt`
- `rlviprev_word`
- `expandstringunsplit`
- `ifs_var`
- `rlkeymap`
- `de_backslash`
- `bashdirectorycompletion_matches`
- `rlexecutingkeymap`
- `rlcompletewithtildeexpansion`
- `same_file`
- `_datastart`
- `currentcommandfirstlinesaved`
- `bashhistorydisable`
- `whichsetflags`
- `rldisplaymatch_list`
- `rldeletehorizontal_space`
- `rlunbindkeyinmap`
- `rlmovecursorrelative`
- `getgrouplist`
- `bashinitializehistory`
- `sh_mktmpfp`
- `globignorecase`
- `allshellvariables`
- `listonejob`
- `rlreplacefrom_history`
- `sh_invalidid`
- `rlviput`
- `rlbackwardkill_line`
- `rlfreeundo_list`
- `zreadintr`
- `historydelimitingchars`
- `withoutjobcontrol`
- `rlarrowkeys`
- `makecommandstring`
- `type_doc`
- `map_over`
- `setdefaultlang`
- `disposeusedenv_vars`
- `lcurrent`
- `fifos_pending`
- `rlcallbackdatadispose`
- `makeifcommand`

- wait_for
- variable~~help~~doc
- this~~command~~name
- source_file
- ignore~~tty~~job_signals
- clean_itemlist
- subshell_envp
- sv_histsize
- history_doc
- r~~free~~historyentry
- close~~fd~~bitmap
- bash~~getc~~wderstr
- push~~dollar~~vars
- err_readonly
- rl~~free~~line_state
- set~~working~~directory
- subshell_environment
- rl_outstream
- print~~simple~~command
- sv~~comp~~wordbreaks
- phash_insert
- rl~~save~~prompt
- rl~~dump~~macros
- make~~variable~~value
- read~~from~~stdin
- initialize_signals
- until_doc
- it_functions
- tilde~~expand~~word
- legal~~alias~~name
- fun~~map~~programs~~specific~~entrystart
- expand~~word~~sshellexp
- sh_neednumarg
- tilde~~additional~~prefixes
- command_substitute
- add_alias
- get~~word~~from_string
- get~~host~~namelist
- yychar
- asynchronous_notification
- rl~~completion~~found_quote
- ifs_cmap
- case_doc
- assign~~array~~from_string
- rl~~pre~~input_hook
- shopt_builtin
- dirs_doc
- \$global\$
- getifs
- rl~~char~~searchinternal
- strlist_resize

- *rlvibacktoindent*
- *rlvcheck*
- *executable_file*
- *malloctraceat_exit*
- *initialize_flags*
- *strvecfromword_list*
- *stdin_redirects*
- *bindkeyseqtounixcommand*
- *shbackslashquote*
- *posixly_correct*
- *setloginshell*
- *terminatestoppedjobs*
- *bashbadsuberrmsg*
- *arrayunshiftelement*
- *rlvdelete*
- *rlvitextmodcommand*
- *rlvibWord*
- *rlresetline_state*
- *rlvariabledumper*
- *it_builtins*
- *hist_verify*
- *getnamefor_error*
- *privileged_mode*
- *nolineediting*
- *historycommentchar*
- *rlalreadyprompted*
- *splitatdelims*
- *rlarg_callback*
- *sh_restricted*
- *interrupt_state*
- *thecurrentmaintainer*
- *rlttysetdefaultbindings*
- *sh_setlinebuf*
- *rlunixline_discard*
- *rlvifWord*
- *rlmakekeymap*
- *rlfix_point*
- *rlwithmacroinput*
- *shbackslashquotefordouble_quotes*
- *validarrayreference*
- *sh_makepath*
- *rlfindprevmbchar*
- *expandwordsno_vars*
- *rlcompletionentry_function*
- *rlvinsert_beg*
- *fmtulong*
- *long_args*
- *pipefail_opt*
- *mailstat*
- *historysearchdelimiter_chars*
- *historysearchpos*

- `mapoverfuncs`
- `rlvchar_search`
- `sh_nojobs`
- `ttstore`
- `printcondcommand`
- `rlttysetdefault_bindings`
- `assignarrayvarfromword_list`
- `rlbeginundo_group`
- `rlinitializemap`
- `rllibraryversion`
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- GLIBC_2.0
- GLIBC_2.4
- GLIBC_2.3
- GLIBC_2.3.4
- GLIBC_2.2
- `?m+d0
- `6]+b0
- ` @+b0
- Z @+b0
- |@ 6|=
- D 8+d0
- @ +b0
- |
- bA(+b0
- |`+b0
- t@B+b0
- |+<+b0
- b@(+b0
- b@(+b0
- |7 +b0
- |)p+b0
- b@(+b0
- `:=4|?
- |0h+b0
- \0@+b0
- `!X+b0
- ` (+b0
- b@ +b0
- b@(+b0
- b@(+b0
- <x+b0
- b@ +b0
- `<:+b0
- | p+b0
- |Rr+b0
- `1%+b0
- b@(+b0
- b@(+b0
- b@(+b0
- b@ +b0
- b@(+b0
- b@(+b0
- b@(+b0
- b@(+b0
- b@(+b0
- |%7+b0
- b@(+b0
- `4Y+b0
- b@(+b0
- bl@+b0
- b@(+b0
- b@(+b0
- b@(+b0

- b@(+b0
- `.=+b0
- b@(+b0
- b@(+b0
- |%`+d0
- |p+b0
- |"@+d0
- 0Hd?
- 0Hd?
- 0Hd?
- 0Hd?
- @ @+`
- |@`4|=
- `9%+b0
- |# 6y?
- `X+b0
- `4/+b0
- ``X+c
- _ 0+c
- _ 0+c
- | +c
- `!x+b0
- \ (+d0
- |6`6|?
- `!h+d0
- `:04c?
- ?}6s?3
- #J4c?
- >m+b0
- `\$H+b0
- +d0
- b x+b0
- B+b
- |b +c
- | (+b0
- `x+d0
- 06u?S6u?
- |% 4y?
- |?u+`
- 06t?S6t?
- 06t?S6t?
- %0+
- `?g7Z?
- |h6t??4
- Z@p+`
- R+
- 06u?S6u?
- \A 7\?
- |85+a0
- E:m+b0
- @/M+a0
- `@p+a0

- `??7Z?
- @+`
- ?979(&k
- \R4c?
- @`0+d0
- `X+d0
- \0:7\?
- |-b6t?
- #(+a0
- ! +a0
- |#X4|?
- ?\$j7!?
- 9 B4=?
- 9 J4=?
- /bin/sh
- I have no name!
- BASH_ENV
- GNU bash, version %s-(%s)
- hppa2.0-unknown-linux-gnu
- Usage:
- %s [GNU long option] [option] ...
- %s [GNU long option] [option] script-file ...
- GNU long options:
- Shell options:
- -irsD or -c command or -O shopt_option
- (invocation only)
- -%s or -o option
- Type `%s -c "help set"' for more information about shell options.
- Type `%s -c help' for more information about shell builtin commands.
- Use the `bashbug' command to report bugs.
- /usr/local/share/bashdb/bashdb-main.inc
- run_wordexp
- --wordexp
- runonerecommand
- FUNCNAME
- BASH_SOURCE
- BASH_LINENO
- %s: is a directory
- %s: cannot execute binary file
- POSIXLY_CORRECT
- POSIX_PEDANTIC
- \s-\w\$
- ~/.bashrc
- %s: option requires an argument
- %s: invalid option
- %c%c: invalid option
- ??host??
- SSH_CLIENT
- SSH2_CLIENT
- /etc/bash/bashrc
- /etc/profile
- ~/.profile

- ~/.bash_profile
- ~/.bash_login
- debugger
- dump-po-strings
- dump-strings
- noediting
- noprofile
- protected
- rcfile
- verbose
- version
- timed out waiting for input: auto-logout
- PROMPT_COMMAND
- reader_loop
- PIPESTATUS
- readline stdin
- syntax error near unexpected token `%s'
- syntax error near `%s'
- syntax error: unexpected end of file
- syntax error
- %a %b %d
- %H:%M:%S
- %l:%M:%S
- %l:%M %p
- unexpected EOF while looking for matching `%c'
- unexpected EOF while looking for `]]'
- syntax error in conditional expression: unexpected token `%s'
- syntax error in conditional expression
- unexpected EOF while looking for matching `)'
- unexpected token `s', expected `)'
- expected `)'
- unexpected argument `%s' to conditional unary operator
- unexpected argument to conditional unary operator
- unexpected token `%s', conditional binary operator expected
- conditional binary operator expected
- unexpected argument `%s' to conditional binary operator
- unexpected argument to conditional binary operator
- unexpected token `%c' in conditional command
- unexpected token `%s' in conditional command
- unexpected token %d in conditional command
- Use "%s" to leave the shell.
- memory exhausted
- select
- function
- newline
- !"#%&'()-
- .1;5207689:
- &,) /vw
- {{{{{{
- FG{{{
- YpqrstPfa
- {IRX{{{gO

- [illegible]

- execute_command
- function_calling
- simple-command
- auto_resume
- substring
- saved_redirects
- saved_redirects
- %s: restricted: cannot specify '/' in command names
- %s: command not found
- execute-shell-function
- real %2R
- user %2U
- sys %2S
- TIMEFORMAT: '%c': invalid format character
- rpopvarcontext: head of shell_variables not a function context
- popvarcontext: no global_variables context
- popscope: *head of shell/variables* not a temporary environment scope
- BASH_ARGV
- BASH_ARGC
- all/local/variables: no function context at current scope
- make/local/variable: no function context at current scope
- OPTERR
- ignorespace
- erasedups
- ignoredups
- ignoreboth
- HISTFILE
- shell-init
- shell level (%d) too high, resetting to 1
- error importing function definition for '%s'
- /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
- HOSTTYPE
- hppa2.0
- OSTYPE
- MACHTYPE
- HOSTNAME
- MAILCHECK
- OPTIND
- BASH_VERSION
- BASH_VERSINFO
- BASHEXECUTIONSTRING
- ~/.sh_history
- ~/.bash_history
- IGNOREEOF
- ignoreeof
- HISTCONTROL
- HISTIGNORE
- SECONDS
- BASH_COMMAND
- BASH_SUBSHELL
- RANDOM
- LINENO

- HISTCMD
- COMP_WORDBREAKS
- DIRSTACK
- GROUPS
- GLOBIGNORE
- HISTFILESIZE
- HISTSIZE
- HISTTIMEFORMAT
- HOSTFILE
- LC_ALL
- LC_COLLATE
- LC_CTYPE
- LC_MESSAGES
- LC_NUMERIC
- LC_TIME
- MAILPATH
- TERMCAP
- TERMINFO
- TEXTDOMAIN
- TEXTDOMAINDIR
- histchars
- line
- %s: %s:%s%d:
- last command: %s
- (null)
- Aborting...
- %s: %s: %d
- %s: warning:
- %s: readonly variable
- %s: unbound variable
- bash-maintainers@gnu.org
- unknown command error
- bad command type
- bad connector
- bad jump
- %s%s%s: %s (error token is "%s")
- expression recursion level exceeded
- syntax error in expression
- recursion stack underflow
- invalid arithmetic base
- value too great for base
- */%+&^|
- syntax error: invalid arithmetic operator
- syntax error: operand expected
- identifier expected after pre-increment or pre-decrement
- missing `)'
- exponent less than 0
- division by 0
- `:' expected for conditional expression
- attempted assignment to non-variable
- bug: bad expassign token
- (core dumped)

- (wd: %s)
- Signal %d
-
- [%d]%c
- Stopped
- Stopped(%s)
- Running
- Done(%d)
- Exit %d
- Unknown status
- %s: line %d:
- (core dumped)
- (wd now: %s)
- notifyofjob_status
- SIGCHLD trap
- [%d] %ld
- describe_pid: %ld: no such pid
- [%ld: %d] tcsetattr
- initializejobcontrol: getpgrp failed
- initializejobcontrol: setpgid
- no job control in this shell
- deleting stopped job %d with process group %ld
- wait_for: No record of process %ld
- %s: job has terminated
- %s: job %d already in background
- [%d]%s
- (wd: %s)
- waitforjob: job %d is stopped
- wait: pid %ld is not a child of this shell
- job-working-directory
- start_pipeline: pgrp pipe
- child setpgid (%ld to %ld)
- forked pid %d appears in running job %d
- cannot make pipe for command substitution
- cannot make child for command substitution
- command_substitute: cannot duplicate pipe as fd 1
- command substitution
- %s: cannot assign list to array member
- bad substitution: no closing `%' in %s
- #%-:=?+|}
- %s: substring expression < 0
- %s: bad substitution
- \$%s: cannot assign in this way
- cannot make pipe for process substitution
- /dev/fd/
- cannot make child for process substitution
- cannot duplicate named pipe %s as fd %d
- process substitution
- bad substitution: no closing "" in %s
- no match: %s
- %s: parameter null or not set
- You have mail in \$_

- You have new mail in \$_
- The mail in %s has been read
- interrupt trap
- return trap
- error trap
- debug trap
- exit trap
- invalid signal number
- *runpendingtraps*: bad value in trap_list[%d]: %p
- *runpendingtraps*: signal handler is SIG_DFL, resending %d (%s) to myself
- trap_handler: bad signal %d
- SIGHUP
- SIGINT
- SIGQUIT
- SIGILL
- SIGTRAP
- SIGABRT
- SIGEMT
- SIGFPE
- SIGKILL
- SIGBUS
- SIGSEGV
- SIGSYS
- SIGPIPE
- SIGALRM
- SIGTERM
- SIGUSR1
- SIGUSR2
- SIGCHLD
- SIGPWR
- SIGVTALRM
- SIGPROF
- SIGWINCH
- SIGSTOP
- SIGTSTP
- SIGCONT
- SIGTTIN
- SIGTTOU
- SIGURG
- SIGLOST
- SIGJUNK(31)
- SIGJUNK(32)
- SIGXCPU
- SIGXFSZ
- SIGJUNK(35)
- SIGSTKFLT
- SIGJUNK(37)
- SIGJUNK(38)
- SIGRTMIN
- SIGRTMIN+1
- SIGRTMIN+2
- SIGRTMIN+3

- SIGRTMIN+4
- SIGRTMIN+5
- SIGRTMIN+6
- SIGRTMIN+7
- SIGRTMIN+8
- SIGRTMIN+9
- SIGRTMIN+10
- SIGRTMIN+11
- SIGRTMIN+12
- SIGRTMAX-12
- SIGRTMAX-11
- SIGRTMAX-10
- SIGRTMAX-9
- SIGRTMAX-8
- SIGRTMAX-7
- SIGRTMAX-6
- SIGRTMAX-5
- SIGRTMAX-4
- SIGRTMAX-3
- SIGRTMAX-2
- SIGRTMAX-1
- SIGRTMAX
- RETURN
- cannot allocate new file descriptor for bash input from fd %d
- save**bash**input: buffer already exists for new fd %d
- %s: integer expression expected
- argument expected
- %s: unary operator expected
- %s: binary operator expected
- `)' expected
- `)' expected, found %s
- missing `]'
- too many arguments
- %s.%d(%d)-%s
- %s.%d(%d)
- GNU bash, version %s (%s)
- Copyright (C) 2005 Free Software Foundation, Inc.
- release
- @(#)Bash version 3.2.17(1) release GNU
- %s[%s: %s
- [%s]: %s
- %s: cannot assign to non-numeric index
- array assign
- bad array subscript
- b%s%s%s
- ;&()|<>
- %s: cannot create: %s
- R%s: first non-whitespace character is not `"'
- no closing `%c' in %s
- %s: missing colon separator
- bashexecuteunix_command: cannot find keymap for command
- bashexecuteunix_command

- completion-ignore-case
- ;|&{(`
- \$include
- hostname*completion*file
- /etc/hosts
- symlink-hook
- (<>|
- C-xC-e
- fc -e "\${VISUAL:-\${EDITOR:-emacs}}"
- fc -e vi
- fc -e "\${VISUAL:-\${EDITOR:-vi}}"
- shell-expand-line
- history-expand-line
- magic-space
- history-and-alias-expand-line
- insert-last-argument
- operate-and-get-next
- display-shell-version
- edit-and-execute-command
- complete-into-braces
- complete-filename
- possible-filename-completions
- complete-username
- possible-username-completions
- complete-hostname
- possible-hostname-completions
- complete-variable
- possible-variable-completions
- complete-command
- possible-command-completions
- glob-complete-word
- glob-expand-word
- glob-list-expansions
- dynamic-complete-history
- \"@<>=;|&()#`?\*`![:{
- comment-begin
- FIGNORE
- \"@><=;|&(:
- \"><=;|&(:
- ,#: %s:%d
- msgid %s%s
- msgstr \"
- /usr/share/locale
- redirection error: cannot duplicate fd
- sh-thd
- file descriptor out of range
- %s: ambiguous redirect
- %s: cannot overwrite existing file
- %s: restricted: cannot redirect output
- cannot create temp file for here document: %s
- /dev/tcp//
- /dev/udp//

- COMP_LINE
- COMP_POINT
- COMP_WORDS
- COMP_CWORD
- completion: function `%s' not found
- COMPREPLY
- jprocomp_insert: %s: NULL COMPSPEC
- xrealloc: cannot reallocate %lu bytes (%lu bytes allocated)
- xmalloc: cannot allocate %lu bytes (%lu bytes allocated)
- alias [-p] [name[=value] ...]
- unalias
- unalias [-a] name [name ...]
- bind [-lpvsPVS] [-m keymap] [-f filename] [-q name] [-u name] [-r keyseq] [-x keyseq:shell-command]
- break [n]
- continue
- continue [n]
- builtin [shell-builtin [arg ...]]
- caller
- caller [EXPR]
- cd [-L|-P] [dir]
- pwd [-LP]
- command [-pVv] command [arg ...]
- declare
- declare [-afFrtx] [-p] [name[=value] ...]
- typeset
- typeset [-afFrtx] [-p] name[=value] ...
- local name[=value] ...
- echo [-neE] [arg ...]
- enable
- enable [-pnds] [-a] [-f filename] [name ...]
- eval [arg ...]
- getopts
- getopts optstring name [arg]
- exec [-cl] [-a name] file [redirection ...]
- exit [n]
- fc [-e ename] [-nlr] [first] [last] or fc -s [pat=rep] [cmd]
- fg [job_spec]
- bg [job_spec ...]
- hash [-lr] [-p pathname] [-dt] [name ...]
- help [-s] [pattern ...]
- history [-c] [-d offset] [n] or history -awrn [filename] or history -ps arg [arg...]
- jobs [-lnprs] [jobspec ...] or jobs -x command [args]
- disown
- disown [-h] [-ar] [jobspec ...]
- kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
- let arg [arg ...]
- read [-ers] [-u fd] [-t timeout] [-p prompt] [-a array] [-n nchars] [-d delim] [name ...]
- return
- return [n]
- set [--abefhkmnptuvxBCHP] [-o option] [arg ...]
- unset [-f] [-v] [name ...]
- export

- export [-nf] [name[=value] ...] or export -p
- readonly
- readonly [-af] [name[=value] ...] or readonly -p
- shift [n]
- source
- source filename [arguments]
- . filename [arguments]
- suspend
- suspend [-f]
- test [expr]
- [arg...]
- trap [-lp] [arg signal_spec ...]
- type [-afptP] name [name ...]
- ulimit
- ulimit [-SHacdefilmnpqrstuvx] [limit]
- umask [-p] [-S] [mode]
- wait [n]
- for NAME [in WORDS ... ;] do COMMANDS; done
- for ((exp1; exp2; exp3)); do COMMANDS; done
- select NAME [in WORDS ... ;] do COMMANDS; done
- time [-p] PIPELINE
- case WORD in [PATTERN [| PATTERN]...] COMMANDS ;;]... esac
- if COMMANDS; then COMMANDS; [elif COMMANDS; then COMMANDS;]... [else COMMANDS;] fi
- while COMMANDS; do COMMANDS; done
- until COMMANDS; do COMMANDS; done
- function NAME { COMMANDS ; } or NAME () { COMMANDS ; }
- { ... }
- JOB_SPEC [&]
- ((...))
- ((expression))
- [[...]]
- [[expression]]
- variables
- variables - Some variable names and meanings
- pushd [dir | +N | -N] [-n]
- popd [+N | -N] [-n]
- dirs [-clpv] [+N] [-N]
- shopt [-pqsu] [-o long-option] optname [optname...]
- printf
- printf [-v var] format [arguments]
- complete [-abcdefgjksuv] [-pr] [-o option] [-A action] [-G globpat] [-W wordlist] [-P prefix] [-S su
- compgen
- compgen [-abcdefgjksuv] [-o option] [-A action] [-G globpat] [-W wordlist] [-P prefix] [-S suffix] [
- `alias' with no arguments or with the -p option prints the list
- of aliases in the form alias NAME=VALUE on standard output.
- Otherwise, an alias is defined for each NAME whose VALUE is given.
- A trailing space in VALUE causes the next word to be checked for
- alias substitution when the alias is expanded. Alias returns
- true unless a NAME is given for which no alias has been defined.
- Remove NAMEs from the list of defined aliases. If the -a option is given,
- then remove all alias definitions.
- Bind a key sequence to a Readline function or a macro, or set

- a Readline variable. The non-option argument syntax is equivalent
- to that found in ~/.inputrc, but must be passed as a single argument:
- bind "'\C-x\C-r': re-read-init-file'.
- bind accepts the following options:
- -m keymap Use 'keymap' as the keymap for the duration of this
- command. Acceptable keymap names are emacs,
- emacs-standard, emacs-meta, emacs-ctlx, vi, vi-move,
- vi-command, and vi-insert.
- -l List names of functions.
- -P List function names and bindings.
- -p List functions and bindings in a form that can be
- reused as input.
- -r keyseq Remove the binding for KEYSEQ.
- -x keyseq:shell-command
- Cause SHELL-COMMAND to be executed when
- KEYSEQ is entered.
- -f filename Read key bindings from FILENAME.
- -q function-name Query about which keys invoke the named function.
- -u function-name Unbind all keys which are bound to the named function.
- -V List variable names and values
- -v List variable names and values in a form that can
- be reused as input.
- -S List key sequences that invoke macros and their values
- -s List key sequences that invoke macros and their values
- in a form that can be reused as input.
- Exit from within a FOR, WHILE or UNTIL loop. If N is specified,
- break N levels.
- Resume the next iteration of the enclosing FOR, WHILE or UNTIL loop.
- If N is specified, resume at the N-th enclosing loop.
- Run a shell builtin. This is useful when you wish to rename a
- shell builtin to be a function, but need the functionality of the
- builtin within the function itself.
- Returns the context of the current subroutine call.
- Without EXPR, returns "\$line \$filename". With EXPR,
- returns "\$line \$subroutine \$filename"; this extra information
- can be used to provide a stack trace.
- The value of EXPR indicates how many call frames to go back before the
- current one; the top frame is frame 0.
- Change the current directory to DIR. The variable \$HOME is the
- default DIR. The variable CDPATH defines the search path for
- the directory containing DIR. Alternative directory names in CDPATH
- are separated by a colon (:). A null directory name is the same as
- the current directory, i.e. `.`. If DIR begins with a slash (/),
- then CDPATH is not used. If the directory is not found, and the
- shell option `cdable_vars' is set, then try the word as a variable
- name. If that variable has a value, then cd to the value of that
- variable. The -P option says to use the physical directory structure
- instead of following symbolic links; the -L option forces symbolic links
- to be followed.
- Print the current working directory. With the -P option, pwd prints
- the physical directory, without any symbolic links; the -L option
- makes pwd follow symbolic links.

- No effect; the command does nothing. A zero exit code is returned.
- Return a successful result.
- Return an unsuccessful result.
- Runs COMMAND with ARGS ignoring shell functions. If you have a shell function called `ls`, and you wish to call the command `ls`, you can say "command `ls`". If the `-p` option is given, a default value is used for PATH that is guaranteed to find all of the standard utilities. If the `-V` or `-v` option is given, a string is printed describing COMMAND.
- The `-V` option produces a more verbose description.
- Declare variables and/or give them attributes. If no NAMES are given, then display the values of variables instead. The `-p` option will display the attributes and values of each NAME.
- The flags are:
 - `-a`
 - to make NAMES arrays (if supported)
 - `-f`
 - to select from among function names only
 - `-F`
 - to display function names (and line number and source file name if debugging) without definitions
 - `-i`
 - to make NAMES have the 'integer' attribute
 - `-r`
 - to make NAMES readonly
 - `-t`
 - to make NAMES have the 'trace' attribute
 - `-x`
 - to make NAMES export
- Variables with the integer attribute have arithmetic evaluation (see 'let') done when the variable is assigned to.
- When displaying values of variables, `-f` displays a function's name and definition. The `-F` option restricts the display to function name only.
- Using `+' instead of '-'` turns off the given attribute instead. When used in a function, makes NAMES local, as with the 'local' command.
- Obsolete. See 'declare'.
- Create a local variable called NAME, and give it VALUE. LOCAL can only be used within a function; it makes the variable NAME have a visible scope restricted to that function and its children.
- Output the ARGs. If `-n` is specified, the trailing newline is suppressed. If the `-e` option is given, interpretation of the following backslash-escaped characters is turned on:
 - alert (bell)
 - backspace
 - suppress trailing newline
 - escape character
 - form feed
 - new line
 - carriage return
 - horizontal tab
 - vertical tab
 - backslash

- the character whose ASCII code is NNN (octal). NNN can be
- 0 to 3 octal digits
- You can explicitly turn off the interpretation of the above characters
- with the -E option.
- Enable and disable builtin shell commands. This allows
- you to use a disk command which has the same name as a shell
- builtin without specifying a full pathname. If -n is used, the
- NAMEs become disabled; otherwise NAMEs are enabled. For example,
- to use the 'test' found in \$PATH instead of the shell builtin
- version, type 'enable -n test'. On systems supporting dynamic
- loading, the -f option may be used to load new builtins from the
- shared object FILENAME. The -d option will delete a builtin
- previously loaded with -f. If no non-option names are given, or
- the -p option is supplied, a list of builtins is printed. The
- -a option means to print every builtin with an indication of whether
- or not it is enabled. The -s option restricts the output to the POSIX.2
- 'special' builtins. The -n option displays a list of all disabled builtins.
- Read ARGs as input to the shell and execute the resulting command(s).
- Getopts is used by shell procedures to parse positional parameters.
- OPTSTRING contains the option letters to be recognized; if a letter
- is followed by a colon, the option is expected to have an argument,
- which should be separated from it by white space.
- Each time it is invoked, getopts will place the next option in the
- shell variable \$name, initializing name if it does not exist, and
- the index of the next argument to be processed into the shell
- variable OPTIND. OPTIND is initialized to 1 each time the shell or
- a shell script is invoked. When an option requires an argument,
- getopts places that argument into the shell variable OPTARG.
- getopts reports errors in one of two ways. If the first character
- of OPTSTRING is a colon, getopts uses silent error reporting. In
- this mode, no error messages are printed. If an invalid option is
- seen, getopts places the option character found into OPTARG. If a
- required argument is not found, getopts places a ':' into NAME and
- sets OPTARG to the option character found. If getopts is not in
- silent mode, and an invalid option is seen, getopts places '?' into
- NAME and unsets OPTARG. If a required argument is not found, a '?'
- is placed in NAME, OPTARG is unset, and a diagnostic message is
- printed.
- If the shell variable OPTERR has the value 0, getopts disables the
- printing of error messages, even if the first character of
- OPTSTRING is not a colon. OPTERR has the value 1 by default.
- Getopts normally parses the positional parameters (\$0 - \$9), but if
- more arguments are given, they are parsed instead.
- Exec FILE, replacing this shell with the specified program.
- If FILE is not specified, the redirections take effect in this
- shell. If the first argument is '-', then place a dash in the
- zeroth arg passed to FILE, as login does. If the '-c' option
- is supplied, FILE is executed with a null environment. The '-a'
- option means to make set argv[0] of the executed process to NAME.
- If the file cannot be executed and the shell is not interactive,
- then the shell exits, unless the shell option 'execfail' is set.
- Exit the shell with a status of N. If N is omitted, the exit status

- is that of the last command executed.
- Logout of a login shell.
- `fc` is used to list or edit and re-execute commands from the history list.
- `FIRST` and `LAST` can be numbers specifying the range, or `FIRST` can be a string, which means the most recent command beginning with that string.
- `-e ENAME` selects which editor to use. Default is `FCEDIT`, then `EDITOR`, then `vi`.
- `-l` means list lines instead of editing.
- `-n` means no line numbers listed.
- `-r` means reverse the order of the lines (making it newest listed first).
- With the ``fc -s [pat=rep ...] [command]`` format, the command is re-executed after the substitution `OLD=NEW` is performed.
- A useful alias to use with this is `r='fc -s'`, so that typing ``r cc'` runs the last command beginning with `cc'` and typing `r'` re-executes the last command.
- Place `JOB_SPEC` in the foreground, and make it the current job. If `JOB_SPEC` is not present, the shell's notion of the current job is used.
- Place each `JOB_SPEC` in the background, as if it had been started with ``&'`. If `JOB_SPEC` is not present, the shell's notion of the current job is used.
- For each `NAME`, the full pathname of the command is determined and remembered. If the `-p` option is supplied, `PATHNAME` is used as the full pathname of `NAME`, and no path search is performed. The `-r` option causes the shell to forget all remembered locations. The `-d` option causes the shell to forget the remembered location of each `NAME`.
- If the `-t` option is supplied the full pathname to which each `NAME` corresponds is printed. If multiple `NAME` arguments are supplied with `-t`, the `NAME` is printed before the hashed full pathname. The `-l` option causes output to be displayed in a format that may be reused as input.
- If no arguments are given, information about remembered commands is displayed.
- Display helpful information about builtin commands. If `PATTERN` is specified, gives detailed help on all commands matching `PATTERN`, otherwise a list of the builtins is printed. The `-s` option restricts the output for each builtin command matching `PATTERN` to a short usage synopsis.
- Display the history list with line numbers. Lines listed with `with a '*'` have been modified. Argument of `N` says to list only the last `N` lines. The ``-c'` option causes the history list to be cleared by deleting all of the entries. The ``-d'` option deletes the history entry at offset `OFFSET`. The ``-w'` option writes out the current history to the history file; ``-r'` means to read the file and append the contents to the history list instead. ``-a'` means to append history lines from this session to the history file.
- Argument ``-n'` means to read all history lines not already read from the history file and append them to the history list.
- If `FILENAME` is given, then that is used as the history file else if `$HISTFILE` has a value, that is used, else `~/.bash_history`.
- If the `-s` option is supplied, the non-option `ARGS` are appended to the history list as a single entry. The `-p` option means to perform history expansion on each `ARG` and display the result, without storing

- anything in the history list.
- If the \$HISTTIMEFORMAT variable is set and not null, its value is used
- as a format string for strftime(3) to print the time stamp associated
- with each displayed history entry. No time stamps are printed otherwise.
- Lists the active jobs. The -l option lists process id's in addition
- to the normal information; the -p option lists process id's only.
- If -n is given, only processes that have changed status since the last
- notification are printed. JOBSPEC restricts output to that job. The
- -r and -s options restrict output to running and stopped jobs only,
- respectively. Without options, the status of all active jobs is
- printed. If -x is given, COMMAND is run after all job specifications
- that appear in ARGS have been replaced with the process ID of that job's
- process group leader.
- By default, removes each JOBSPEC argument from the table of active jobs.
- If the -h option is given, the job is not removed from the table, but is
- marked so that SIGHUP is not sent to the job if the shell receives a
- SIGHUP. The -a option, when JOBSPEC is not supplied, means to remove all
- jobs from the job table; the -r option means to remove only running jobs.
- Send the processes named by PID (or JOBSPEC) the signal SIGSPEC. If
- SIGSPEC is not present, then SIGTERM is assumed. An argument of '-'
- lists the signal names; if arguments follow '-' they are assumed to
- be signal numbers for which names should be listed. Kill is a shell
- builtin for two reasons: it allows job IDs to be used instead of
- process IDs, and, if you have reached the limit on processes that
- you can create, you don't have to start a process to kill another one.
- Each ARG is an arithmetic expression to be evaluated. Evaluation
- is done in fixed-width integers with no check for overflow, though
- division by 0 is trapped and flagged as an error. The following
- list of operators is grouped into levels of equal-precedence operators.
- The levels are listed in order of decreasing precedence.
- id++, id--
- variable post-increment, post-decrement
- ++id, --id
- variable pre-increment, pre-decrement
- unary minus, plus
- logical and bitwise negation
- exponentiation
- *, /, %
- multiplication, division, remainder
- addition, subtraction
- <<, >>
- left and right bitwise shifts
- <=, >=, <, >
- comparison
- ==, !=
- equality, inequality
- bitwise AND
- bitwise XOR
- bitwise OR
- logical AND
- logical OR
- expr ? expr : expr

- conditional operator
- =, *=, /=, %=,
- +=, -=, <=<, >>=,
- &=, ^=, |=
- assignment
- Shell variables are allowed as operands. The name of the variable
- is replaced by its value (coerced to a fixed-width integer) within
- an expression. The variable need not have its integer attribute
- turned on to be used in an expression.
- Operators are evaluated in order of precedence. Sub-expressions in
- parentheses are evaluated first and may override the precedence
- rules above.
- If the last ARG evaluates to 0, let returns 1; 0 is returned
- otherwise.
- One line is read from the standard input, or from file descriptor FD if the
- -u option is supplied, and the first word is assigned to the first NAME,
- the second word to the second NAME, and so on, with leftover words assigned
- to the last NAME. Only the characters found in \$IFS are recognized as word
- delimiters. If no NAMEs are supplied, the line read is stored in the REPLY
- variable. If the -r option is given, this signifies 'raw' input, and
- backslash escaping is disabled. The -d option causes read to continue
- until the first character of DELIM is read, rather than newline. If the -p
- option is supplied, the string PROMPT is output without a trailing newline
- before attempting to read. If -a is supplied, the words read are assigned
- to sequential indices of ARRAY, starting at zero. If -e is supplied and
- the shell is interactive, readline is used to obtain the line. If -n is
- supplied with a non-zero NCHARS argument, read returns after NCHARS
- characters have been read. The -s option causes input coming from a
- terminal to not be echoed.
- The -t option causes read to time out and return failure if a complete line
- of input is not read within TIMEOUT seconds. If the TMOUT variable is set,
- its value is the default timeout. The return code is zero, unless end-of-file
- is encountered, read times out, or an invalid file descriptor is supplied as
- the argument to -u.
- Causes a function to exit with the return value specified by N. If N
- is omitted, the return status is that of the last command.
- -a Mark variables which are modified or created for export.
- -b Notify of job termination immediately.
- -e Exit immediately if a command exits with a non-zero status.
- -f Disable file name generation (globbing).
- -h Remember the location of commands as they are looked up.
- -k All assignment arguments are placed in the environment for a
- command, not just those that precede the command name.
- -m Job control is enabled.
- -n Read commands but do not execute them.
- -o option-name
- Set the variable corresponding to option-name:
- allexport same as -a
- braceexpand same as -B
- emacs use an emacs-style line editing interface
- errexit same as -e
- errtrace same as -E

- functrace same as -T
- hashall same as -h
- histexpand same as -H
- history enable command history
- ignoreeof the shell will not exit upon reading EOF
- interactive-comments
- allow comments to appear in interactive commands
- keyword same as -k
- monitor same as -m
- noclobber same as -C
- noexec same as -n
- noglob same as -f
- nolog currently accepted but ignored
- notify same as -b
- nounset same as -u
- onecmd same as -t
- physical same as -P
- pipefail the return value of a pipeline is the status of
- the last command to exit with a non-zero status,
- or zero if no command exited with a non-zero status
- posix change the behavior of bash where the default
- operation differs from the 1003.2 standard to
- match the standard
- privileged same as -p
- verbose same as -v
- vi use a vi-style line editing interface
- xtrace same as -x
- -p Turned on whenever the real and effective user ids do not match.
- Disables processing of the \$ENV file and importing of shell
- functions. Turning this option off causes the effective uid and
- gid to be set to the real uid and gid.
- -t Exit after reading and executing one command.
- -u Treat unset variables as an error when substituting.
- -v Print shell input lines as they are read.
- -x Print commands and their arguments as they are executed.
- -B the shell will perform brace expansion
- -C If set, disallow existing regular files to be overwritten
- by redirection of output.
- -E If set, the ERR trap is inherited by shell functions.
- -H Enable ! style history substitution. This flag is on
- by default when the shell is interactive.
- -P If set, do not follow symbolic links when executing commands
- such as cd which change the current directory.
- -T If set, the DEBUG trap is inherited by shell functions.
- - Assign any remaining arguments to the positional parameters.
- The -x and -v options are turned off.
- Using + rather than - causes these flags to be turned off. The
- flags can also be used upon invocation of the shell. The current
- set of flags may be found in \$-. The remaining n ARGs are positional
- parameters and are assigned, in order, to \$1, \$2, .. \$n. If no

- ARGs are given, all shell variables are printed.
- For each NAME, remove the corresponding variable or function. Given
- the `-v`, `unset` will only act on variables. Given the `-f` flag,
- `unset` will only act on functions. With neither flag, `unset` first
- tries to `unset` a variable, and if that fails, then tries to `unset` a
- function. Some variables cannot be `unset`; also see `readonly`.
- NAMEs are marked for automatic export to the environment of
- subsequently executed commands. If the `-f` option is given,
- the NAMEs refer to functions. If no NAMEs are given, or if `-p` is given, a list of all names that are
- exported in this shell is printed. An argument of `-n` says to remove the export property
- from subsequent NAMEs. An argument of `--` disables further option processing. The given
- NAMEs are marked `readonly` and the values of these NAMEs may not be changed by subsequent
- assignment. If the `-f` option is given, then functions corresponding to the NAMEs are so marked.
- If no arguments are given, or if `-p` is given, a list of all `readonly` names
- is printed. The `-a` option means to treat each NAME as an array variable. An argument of `--` disables
- further option
- processing.
- The positional parameters from `$N+1 ...` are renamed to `$1 ...`. If `N` is
- not given, it is assumed to be 1.
- Read and execute commands from FILENAME and return. The pathnames
- in `$PATH` are used to find the directory containing FILENAME. If any
- ARGUMENTS are supplied, they become the positional parameters when
- FILENAME is executed.
- Suspend the execution of this shell until it receives a `SIGCONT`
- signal. The `-f` if specified says not to complain about this being a login shell if it is; just
- suspend anyway. Exits with a status of 0 (true) or 1 (false) depending on the
- evaluation of EXPR. Expressions may be unary or binary. Unary expressions are often used to
- examine the status of a file. There are string operators as well, and numeric comparison
- operators. File operators:
- `-a FILE` True if file exists.
- `-b FILE` True if file
- is block special.
- `-c FILE` True if file is character special.
- `-d FILE` True if file is a
- directory.
- `-e FILE` True if file exists.
- `-f FILE` True if file exists and is a regular
- file.
- `-g FILE` True if file is set-group-id.
- `-h FILE` True if file is a symbolic link.
- `-L FILE` True if file is a symbolic link.
- `-k FILE` True if file has its 'sticky' bit set.
- `-p FILE` True if file is a named pipe.
- `-r FILE` True if file is readable by you.
- `-s FILE` True if file exists and is not empty.
- `-S FILE` True if file is a socket.
- `-t FD` True if FD is opened on a terminal.
- `-u FILE` True if the file is set-user-id.
- `-w FILE` True if the file is writable by you.
- `-x FILE` True if the file is executable by you.
- `-O FILE` True if the file is effectively owned by you.
- `-G FILE` True if the file is effectively owned by your group.
- `-N FILE` True if the file has been modified since it was last read.
- `FILE1 -nt FILE2` True if file1 is newer than file2 (according to
- modification date).
- `FILE1 -ot FILE2` True if file1 is older than file2.
- `FILE1 -ef FILE2` True if file1 is a hard link to file2.
- String operators:
- `-z STRING` True if string is empty.
- `-n STRING`
- `STRING` True if string is not empty.

- `STRING1 = STRING2`
- True if the strings are equal.
- `STRING1 != STRING2`
- True if the strings are not equal.
- `STRING1 < STRING2`
- True if `STRING1` sorts before `STRING2` lexicographically.
- `STRING1 > STRING2`
- True if `STRING1` sorts after `STRING2` lexicographically.
- Other operators:
- `-o OPTION` True if the shell option `OPTION` is enabled.
- `! EXPR` True if `expr` is false.
- `EXPR1 -a EXPR2` True if both `expr1` AND `expr2` are true.
- `EXPR1 -o EXPR2` True if either `expr1` OR `expr2` is true.
- `arg1 OP arg2` Arithmetic tests. `OP` is one of `-eq`, `-ne`,
- `-lt`, `-le`, `-gt`, or `-ge`.
- Arithmetic binary operators return true if `ARG1` is equal, not-equal,
- less-than, less-than-or-equal, greater-than, or greater-than-or-equal
- than `ARG2`.
- This is a synonym for the "test" builtin, but the last
- argument must be a literal `]`, to match the opening `[`.
- Print the accumulated user and system times for processes run from
- the shell.
- The command `ARG` is to be read and executed when the shell receives
- signal(s) `SIGNALSPEC`. If `ARG` is absent (and a single `SIGNALSPEC`
- is supplied) or `-`, each specified signal is reset to its original value. If `ARG` is the null
- string each `SIGNAL_SPEC` is ignored by the shell and by the commands it invokes. If a `SIGNAL_SPEC`
- is `EXIT (0)` the command `ARG` is executed on exit from the shell. If a `SIGNAL_SPEC` is
- `DEBUG`, `ARG` is executed after every simple command. If the `-p` option
- is supplied then the trap commands associated with each `SIGNALSPEC` are
- displayed. If no arguments are supplied or if only `-p` is given, `trap` prints the list of commands
- associated with each signal. Each `SIGNAL_SPEC` is either a signal name in `<signal.h>` or a signal
- number. Signal names are case insensitive and the `SIG` prefix is optional. `trap -l` prints
- a list of signal names and their corresponding numbers. Note that a
- signal can be sent to the shell with `"kill -signal $$"`.
- For each `NAME`, indicate how it would be interpreted if used as a
- command name.
- If the `-t` option is used, `type` outputs a single word which is one of `<code>alias',</code>keyword',`
- `<code>function',</code>builtin', <code>file' or</code>'`, if `NAME` is an alias, shell reserved
- word, shell function, shell builtin, disk file, or unfound, respectively. If the `-p`
- flag is used, `type` either returns the name of the disk
- file that would be executed, or nothing if `type -t NAME` would not return `file`.
- If the `-a` flag is used, `type` displays all of the places that contain an executable named `file`. This
- includes aliases, builtins, and
- functions, if and only if the `-p` flag is not also used.
- The `-f` flag suppresses shell function lookup.
- The `-P` flag forces a `PATH` search for each `NAME`, even if it is an alias,
- builtin, or function, and returns the name of the disk file that would
- be executed.
- `Ulimit` provides control over the resources available to processes
- started by the shell, on systems that allow such control. If an
- option is given, it is interpreted as follows:
- `-S`

- use the `soft` resource limit
- `-a`
- all current limits are reported
- `-c`
- the maximum size of core files created
- `-d`
- the maximum size of a process's data segment
- `-e`
- the maximum scheduling priority (`nice`)
- `-f` the maximum size of files written by the shell and its children
- `-i` the maximum number of pending signals
- `-l` the maximum size a process may lock into memory
- `-m` the maximum resident set size
- `-n` the maximum number of open file descriptors
- `-p` the pipe buffer size
- `-q` the maximum number of bytes in POSIX message queues
- `-r` the maximum real-time scheduling priority
- `-s` the maximum stack size
- `-t` the maximum amount of cpu time in seconds
- `-u` the maximum number of user processes
- `-v` the size of virtual memory
- `-x` the maximum number of file locks
- If `LIMIT` is given, it is the new value of the specified resource; the special `LIMIT` values `soft`, `hard`, and `unlimited` stand for
- the current soft limit, the current hard limit, and no limit, respectively.
- Otherwise, the current value of the specified resource is printed.
- If no option is given, then `-f` is assumed. Values are in 1024-byte increments, except for `-t`, which is in seconds, `-p`, which is in increments of 512 bytes, and `-u`, which is an unscaled number of processes.
- The user file-creation mask is set to `MODE`. If `MODE` is omitted, or if `-S` is supplied, the current value of the mask is printed. The `-S` option makes the output symbolic; otherwise an octal number is output.
- If `-p` is supplied, and `MODE` is omitted, the output is in a form that may be used as input. If `MODE` begins with a digit, it is interpreted as an octal number, otherwise it is a symbolic mode string like that accepted by `chmod(1)`. Wait for the specified process and report its termination status. If `N` is not given, all currently active child processes are waited for, and the return code is zero. `N` may be a process ID or a job specification; if a job spec is given, all processes in the job's pipeline are waited for. The `for` loop executes a sequence of commands for each member in a
- list of items. If in `WORDS ...;` is not present, then in `"$@"` is assumed. For each element in `WORDS`, `NAME` is set to that element, and the `COMMANDS` are executed.
- Equivalent to
- `(( EXP1 ))`
- `while (( EXP2 )); do`
- `COMMANDS`
- `(( EXP3 ))`
- `EXP1`, `EXP2`, and `EXP3` are arithmetic expressions. If any expression is omitted, it behaves as if it evaluates to 1.
- The `WORDS` are expanded, generating a list of words. The set of expanded words is printed on the standard error, each preceded by a number. If in `WORDS'` is not present, in `"$@"` is assumed. The `PS3` prompt is then displayed and a line read from the standard input. If the line consists of the number corresponding to one of the displayed words, then `NAME` is set to that word. If the line is empty, `WORDS` and the prompt are redisplayed. If EOF is read, the command completes. Any other

- value read causes NAME to be set to null. The line read is saved
- in the variable REPLY. COMMANDS are executed after each selection
- until a break command is executed.
- Execute PIPELINE and print a summary of the real time, user CPU time, and system CPU time spent executing PIPELINE when it terminates.
- The return status is the return status of PIPELINE. The `-p` option prints the timing summary in a slightly different format. This uses the value of the TIMEFORMAT variable as the output format. Selectively execute COMMANDS based upon WORD matching PATTERN. The `|` is used to separate multiple patterns.
- The `if` COMMANDS' list is executed. If its exit status is zero, then the `then` COMMANDS' list is executed. Otherwise, each `elif` COMMANDS' list is executed in turn, and if its exit status is zero, the corresponding `then` COMMANDS' list is executed and the `if` command completes. Otherwise,
- the `else` COMMANDS' list is executed, if present. The exit status of the entire construct is the exit status of the last command executed, or zero if no condition tested true. Expand and execute COMMANDS as long as the final command in the `while` COMMANDS has an exit status of zero.
- Expand and execute COMMANDS as long as the final command in the `until` COMMANDS has an exit status which is not zero. Create a simple command invoked by NAME which runs COMMANDS. Arguments on the command line along with NAME are passed to the function as `$0 .. $n`. Run a set of commands in a group. This is one way to redirect an entire set of commands. Equivalent to the JOB_SPEC argument to the `fg` command. Resume a stopped or background job. JOBSPEC can specify either a job name or a job number. Following JOBSPEC with a `&` places the job in the background, as if the job specification had been supplied as an argument to `bg`.
- The EXPRESSION is evaluated according to the rules for arithmetic evaluation. Equivalent to `"let EXPRESSION"`.
- Returns a status of 0 or 1 depending on the evaluation of the conditional expression EXPRESSION. Expressions are composed of the same primaries used by the `test` builtin, and may be combined using the following operators
- `( EXPRESSION )` Returns the value of EXPRESSION
- `! EXPRESSION` True if EXPRESSION is false; else false
- `EXPR1 && EXPR2` True if both EXPR1 and EXPR2 are true; else false
- `EXPR1 || EXPR2` True if either EXPR1 or EXPR2 is true; else false
- When the `==` and `!=` operators are used, the string to the right of the operator is used as a pattern and pattern matching is performed. The `&&` and `||` operators do not evaluate EXPR2 if EXPR1 is sufficient to determine the expression's value.
- `BASH_VERSION` Version information for this Bash.
- `CDPATH` A colon-separated list of directories to search for directories given as arguments to `cd`.
- `GLOBIGNORE` A colon-separated list of patterns describing filenames to be ignored by pathname expansion.
- `HISTFILE` The name of the file where your command history is stored.
- `HISTFILESIZE` The maximum number of lines this file can contain.
- `HISTSZ` The maximum number of history lines that a running shell can access.
- `HOME` The complete pathname to your login directory.
- `HOSTNAME` The name of the current host.

- `HOSTTYPE`
- The type of CPU this version of Bash is running under.
- `IGNOREEOF`
- Controls the action of the shell on receipt of an EOF
- character as the sole input. If set, then the value
- of it is the number of EOF characters that can be seen
- in a row on an empty line before the shell will exit
- (default 10). When unset, EOF signifies the end of input.
- `MACHTYPE`
- A string describing the current system Bash is running on.
- `MAILCHECK`
- How often, in seconds, Bash checks for newmail.
- `MAILPATH`
- A colon-separated list of filenames which Bash checks
- for newmail.
- `OSTYPE`
- The version of Unix this version of Bash is running on.
- `PATH`
- A colon-separated list of directories to search when
- looking for commands.
- `PROMPTCOMMAND`
- A command to be executed before the printing of each
- primary prompt.
- `PS1`
- The primary prompt string.
- `PS2`
- The secondary prompt string.
- `PWD`
- The full pathname of the current directory.
- `SHELLOPTS`
- A colon-separated list of enabled shell options.
- `TERM`
- The name of the current terminal type.
- `TIMEFORMAT`
- The output format for timing statistics displayed by the
- `time` reserved word.
- `auto_resume`
- Non-null means a command word appearing on a line
- itself is first looked for in the list of currently
- stopped jobs. If found there,
- that job is foregrounded.
- A value of `exact` means that the command word must
- exactly match a command in the list of stopped jobs. A
- value of `substring` means that the command word must
- match a substring of the job. Any other
- value means that
- the command must be a prefix of a stopped job.
- `histchars`
- Characters controlling history expansion and quick
- substitution. The first character is the
- history
- substitution character, usually `!`. The second is
- the quick substitution character, usually `^`. The
- third is the history comment character, usually `#`.
- `HISTIGNORE`
- A colon-separated list of patterns used to decide which
- commands should be saved on the history list.
- Adds a directory to the top of the directory stack, or rotates
- the stack, making the new top of the stack the current working
- directory. With no arguments, exchanges the top two directories.
- `+N`

- Rotates the stack so that the Nth directory (counting
- from the left of the list shown by `dirs'`, starting with zero) is at the top.
- Rotates the stack so that the Nth directory (counting from the right of the list shown by `dirs'`, starting with zero) is at the top.
- -n
- suppress the normal change of directory when adding directories
- to the stack, so only the stack is manipulated.
- dir
- adds DIR to the directory stack at the top, making it the
- new current working directory.
- You can see the directory stack with the `dirs'` command.
- Removes entries from the directory stack. With no arguments,
- removes the top directory from the stack, and `cd's` to the new top directory.
- +N removes the Nth entry counting from the left of the list shown by `dirs'`, starting with zero. For example: `popd +0'`
- removes the first directory, `popd +1'` the second.
- -N
- removes the Nth entry counting from the right of the list
- shown by `dirs'`, starting with zero. For example: `popd -0'`
- removes the last directory, `popd -1'` the next to last.
- -n suppress the normal change of directory when removing directories from the stack, so only the stack is manipulated.
- You can see the directory stack with the `dirs'` command.
- Display the list of currently remembered directories. Directories
- find their way onto the list with the `pushd'` command; you can get back up through the list with the `popd'` command.
- The -l flag specifies that `dirs'` should not print shorthand versions of directories which are relative to your home directory. This means that `~/bin'` might be displayed as `/homes/bfox/bin'`. The -v flag causes `dirs'` to print the directory stack with one entry per line,
- prepending the directory name with its position in the stack. The -p
- flag does the same thing, but the stack position is not prepended.
- The -c flag clears the directory stack by deleting all of the elements.
- +N
- displays the Nth entry counting from the left of the list shown by
- `dirs` when invoked without options, starting with zero.
- -N
- displays the Nth entry counting from the right of the list shown by
- `dirs` when invoked without options, starting with zero.
- Toggle the values of variables controlling optional behavior.
- The -s flag means to enable (set) each OPTNAME; the -u flag
- unsets each OPTNAME. The -q flag suppresses output; the exit
- status indicates whether each OPTNAME is set or unset. The -o
- option restricts the OPTNAMEs to those defined for use with
- `set -o'`. With no options, or with the -p option, a list of all settable options is displayed, with an indication of whether or not each is set.
- printf formats and prints ARGUMENTS under control of the FORMAT. FORMAT is a character string which contains three types of objects: plain characters, which are simply copied to standard output, character escape sequences which are converted and copied to the standard output, and format specifications, each of which causes printing of the next successive argument. In addition to the standard printf(1) formats, %b means to expand backslash escape sequences in the corresponding argument, and %q means to quote the argument in a way that can be reused as shell input.
- If the -v option is supplied, the output is placed into the value of the shell variable VAR rather than being sent to the standard output.
- For each NAME, specify how arguments are to be completed.

completion specifications are printed in a way that allows them to be reused as input. The -r option removes a completion specification for each NAME, or, if no NAMES are supplied, all completion specifications. Display the possible completions depending on the options. Intended to be used from within a shell function generating possible completions. If the optional WORD argument is supplied, matches against WORD are generated. alias %s': invalid

alias name

- *bindbuiltin*
- *lvpsf:q:u:m:r:x:*
- %s': invalid keymap name
- %s: cannot read: %s
- %s': unknown function name
- %s is not bound to any keys.
- %s can be invoked via
- "%s"%s
- %s': cannot unbind
- only meaningful in a `for', while', or until' loop`
- loop count
- %s %s %s
- OLDPWD=
- HOME not set
- OLDPWD not set
- CDPATH
- commandbuiltin
- /bin:/usr/bin:/sbin:/usr/sbin
- R%s: error retrieving current directory: %s: %s
- %s: usage:
- %s: ambiguous job spec
- write error: %s
- %s: not a shell builtin
- %s: restricted
- %s: no job control
- %s: no such job
- %s: %s out of range
- %s': not a pid or valid job spec
- %s: invalid signal specification
- SIGJUNK
- Unknown
- %2d) %s
- %s: invalid number
- %s: invalid option name
- %s: not found
- %s: numeric argument required
- cannot use -f to make functions
- %s %d %s
- %s: cannot destroy array variables in this way
- +afprtxF
- can only be used in a function
- adnpsf:
- enable %s
- enable -n %s
- cannot open shared object %s: %s
- struct
- cannot find %s in shared object %s: %s
- %s: not dynamically loaded
- %s: cannot delete: %s
- %s: not a regular file
- %s: file is too large
- evalfile
- parseandexecutetop
- pedispose
- parseandexecute

- %s: cannot execute: %s
- ~/.bashlogout
- /etc/bash/bashlogout
- There are stopped jobs.
- not login shell: use `exit'` `<li>logout</li>` `<li>${FCEDIT:-${EDITOR:-ed}}</li>` `<li>${FCEDIT:-${EDITOR:-vi}}</li>` `<li>fc builtin</li>` `<li>:e:lnrs</li>` `<li>no command found</li>` `<li>history specification</li>` `<li>bash-fc</li>` `<li>%s: cannot open temp file: %s</li>` `<li>current</li>` `<li>job %d started without job control</li>` `<li>builtin hash -p %s %s</li>` `<li>hashing disabled</li>` `<li>dlp:rt</li>` `<li>command</li>` `<li>%s: hash table empty</li>` `<li>These shell commands are defined internally. Type help' to see this list.`
- Type `help name'` to find out more about the function `name'`.
- Use `info bash'` to find out more about the shell in general.`</li>` `<li>Use man -k' orinfo' to find out more about commands not in this list.</li>` `<li>A star (*) next to a name means that the command is disabled.</li>` `<li>Shell commands matching keywords`
- Shell commands matching keyword `</li>` `<li>%s: cannot open: %s</li>` `<li>no help topics match <code>%s'. Try</code>help help' or <code>man -k %s' or</code>info %s'.</li> <li>%5d%c %s%</li> <li>acd:npsrw</li> <li>cannot use more than one of -anrw</li> <li>%s: history expansion failed</li> <li>history position</li> <li>no other options allowed with -x'`
- lpnxs
- jobsbuiltin
- Unknown error
- (%ld) - %s
- %s: arguments must be process or job IDs
-
- %2d %s
- %2d %s
- dirstack
- directory stack empty
- directory stack index
- no other directory
- readbuiltin
- read error: %d: %s
- %s: invalid timeout specification
- %s: invalid file descriptor specification
- %d: invalid file descriptor: %s
- ersa:d:n:p:tu:
- can only `return'` from a function or sourced script `<li>editing-mode</li>` `<li>cannot simultaneously unset a function and a variable</li>` `<li>%s: cannot unset</li>` `<li>%s: cannot unset: readonly %s</li>` `<li>%s: not an array variable</li>` `<li>SHELLOPTS</li>` `<li>set %co %s</li>` `<li>allexport</li>` `<li>braceexpand</li>` `<li>errexit</li>` `<li>errtrace</li>` `<li>functrace</li>` `<li>hashall</li>` `<li>histexpand</li>` `<li>interactive-comments</li>` `<li>keyword</li>` `<li>monitor</li>` `<li>noclobber</li>` `<li>noexec</li>` `<li>noglob</li>` `<li>notify</li>` `<li>nounset</li>` `<li>onecmd</li>` `<li>physical</li>` `<li>pipefail</li>` `<li>privileged</li>` `<li>xtrace</li>` `<li>(declare -%s </li>` `<li>%s -%s </li>` `<li>%s: not a function</li>` `<li>*shift count</li>` `<li>filename argument required</li>` `<li>%s: file not found</li>` `<li>zcannot suspend</li>` `<li>cannot suspend a login shell</li>` `<li>unknown</li>` `<li>trap -- %s %d</li>` `<li>trap -- %s %s</li>` `<li>%s is aliased to %s'`
- alias %s=%s
- %s is a shell keyword
- %s is a function
- %s is a shell builtin
- %s is %s
- %s is hashed (%s)
- (%s, -%c)

- (-%c)
- %-20s %16s
- unlimited
- %s: cannot get limit: %s
- %c': bad command
- %s: cannot modify limit: %s
- core file size
- blocks
- data seg size
- kbytes
- scheduling priority
- file size
- pending signals
- max locked memory
- max memory size
- open files
- pipe size
- 512 bytes
- POSIX message queues
- real-time priority
- stack size
- cpu time
- seconds
- max user processes
- virtual memory
- file locks
- umask
- %c': invalid symbolic mode operator
- %c': invalid symbolic mode character
- u=%s,g=%s,o=%s
- octal number
- OPTARG
- %s: invalid shell option name
- shopt %s %s
- cannot set and unset shell options simultaneously
- cdable_vars
- cdspell
- checkhash
- checkwinsize
- cmdhist
- dotglob
- execfail
- expand_aliases
- extdebug
- extglob
- extquote
- failglob
- force_fignore
- gnu_errfmt
- histappend
- histreedit
- histverify
- hostcomplete
- huponexit
- interactive_comments
- lithist
- login_shell
- mailwarn
- noemptycmd_completion
- nocaseglob
- nocasematch
- nullglob
- progcomp
- promptvars
- restricted_shell
- shift_verbose
- sourcepath
- xpg_echo
- (missing hex digit for \x
- warning: %s: %s
- #'- + 0
- hjlLtz
- %s': missing format character
- %s: illegal option -- %c
- %s: option requires an argument -- %c
- warning: -F option may not work as you expect
- warning: -C option may not work as you expect
- %s: invalid action name
- abcdefgjkoprsvA:G:W:P:S:X:F:C:
- complete
- -o %s
- bashdefault
- dirnames
- filenames
- nospace
- plusdirs
- -A %s
- arrayvar
- binding
- enabled
- helptopic
- hostname
- running
- setopt
- signal
- %s %s
- %s: no completion specification
- service
- BASHREMATCH
- /dev/std
- %s: bad network path specification
- socket
- connect
- %ldm%d.%03ds

- *shmakepath*
- *shrealpath*
- *TMPDIR*
- */var/tmp*
- */usr/tmp*
- *%s/%s-%lu*
- *b%s/cur*
- *%s/tmp*
- *%s/new*
- *invalid base*
- *0123456789ABCDEF*
- *0123456789abcdef*
- *l|h^0%bB*
- *l|hWw^0bB*
- *l|h^0bB*
- *h|^\$0ftFT;,%wbeWBE]*
- **\AaliCcDdPpYyRrSsXx~*
- *accept-line*
- *arrow-key-prefix*
- *backward-byte*
- *backward-char*
- *backward-kill-line*
- *backward-kill-word*
- *backward-word*
- *beginning-of-history*
- *beginning-of-line*
- *call-last-kbd-macro*
- *capitalize-word*
- *character-search*
- *character-search-backward*
- *clear-screen*
- *copy-backward-word*
- *copy-forward-word*
- *copy-region-as-kill*
- *delete-char*
- *delete-char-or-list*
- *delete-horizontal-space*
- *digit-argument*
- *do-lowercase-version*
- *downcase-word*
- *dump-functions*
- *dump-macros*
- *dump-variables*
- *emacs-editing-mode*
- *end-kbd-macro*
- *end-of-history*
- *end-of-line*
- *exchange-point-and-mark*
- *forward-backward-delete-char*
- *forward-byte*
- *forward-char*
- *forward-word*

- history-search-backward
- history-search-forward
- insert-comment
- insert-completions
- kill-whole-line
- kill-line
- kill-region
- kill-word
- menu-complete
- next-history
- non-incremental-forward-search-history
- non-incremental-reverse-search-history
- non-incremental-forward-search-history-again
- non-incremental-reverse-search-history-again
- overwrite-mode
- possible-completions
- previous-history
- quoted-insert
- re-read-init-file
- redraw-current-line
- revert-line
- self-insert
- set-mark
- start-kbd-macro
- tab-insert
- tilde-expand
- transpose-chars
- transpose-words
- tty-status
- universal-argument
- unix-filename-rubout
- unix-line-discard
- unix-word-rubout
- upcase-word
- yank-last-arg
- yank-nth-arg
- yank-pop
- vi-append-eol
- vi-append-mode
- vi-arg-digit
- vi-back-to-indent
- vi-bWord
- vi-bword
- vi-change-case
- vi-change-char
- vi-change-to
- vi-char-search
- vi-column
- vi-complete
- vi-delete
- vi-delete-to
- vi-eWord

- vi-editing-mode
- vi-end-word
- vi-eof-maybe
- vi-eword
- vi-fWord
- vi-fetch-history
- vi-first-print
- vi-fword
- vi-goto-mark
- vi-insert-beg
- vi-insertion-mode
- vi-match
- vi-movement-mode
- vi-next-word
- vi-overstrike
- vi-overstrike-delete
- vi-prev-word
- vi-put
- vi-redo
- vi-replace
- vi-rubout
- vi-search
- vi-search-again
- vi-set-mark
- vi-subst
- vi-tilde-expand
- vi-yank-arg
- vi-yank-to
- --More--
- Display all %d possibilities? (y or n)
- readline: bad value %d for whattodo in rlcomplete
- "\ @\${>}<=<|&{ (

- *readline: debug: insertsomechars: count (%d) != col (%d)*
- */=~.#\$*
- *Rreadline: readlinecallbackreadchar() called with no handler!*
- *(arg: %d)*
- *<>;&|\$*
- *no previous substitution*
- *unknown expansion error*
- *event not found*
- *bad word specifier*
- *substitution failed*
- *unrecognized history modifier*
- *;&()|<>*
- *b.history*
- *xdigit*
- *backspace*
- *vertical-tab*
- *form-feed*
- *carriage-return*
- *exclamation-mark*
- *quotation-mark*
- *number-sign*
- *dollar-sign*
- *percent-sign*
- *ampersand*
- *apostrophe*
- *left-parenthesis*
- *right-parenthesis*
- *asterisk*
- *plus-sign*
- *hyphen*
- *hyphen-minus*
- *period*
- *full-stop*
- *semicolon*
- *less-than-sign*
- *equals-sign*
- *greater-than-sign*
- *question-mark*
- *commercial-at*
- *left-square-bracket*
- *backslash*
- *reverse-solidus*
- *right-square-bracket*
- *circumflex*
- *circumflex-accent*
- *underscore*
- *grave-accent*
- *left-brace*
- *left-curly-bracket*
- *vertical-line*
- *right-brace*
- *right-curly-bracket*

- *.shstrtab*
- *.interp*
- *.note.ABI-tag*
- *.gnu.hash*
- *.dynsym*
- *.dynstr*
- *.gnu.version*
- *.gnu.versionr*
- *.rela.dyn*
- *.rela.plt*
- *.rodata*
- *.PARISC.unwind*
- *.ehframehdr*
- *.eh_frame*
- *.ctors*
- *.dtors*
- *.dynamic*

UTF-8 Strings

No detected.

UTF-16LE Strings

- 244.,,
- &'()55
- " !"# \$3&'()
- /0<=@b1
-]^T9Uwy:~sz
- mnogh_
- !c"# \$%&'()
- 56tufp7/

Extracted URLs

No detected.

Base64 Strings

No valid Base64-encoded strings found.

File Entropy Analysis

- **Entropy:** 6.64
- **Status:** This file is likely packed or encrypted.

VirusTotal Analysis

Scan Date: 2024-12-10 07:32:37

Scan Results

- Malicious: 0

- Suspicious: 0
- Undetected: 64
- Harmless: 0
- Timeout: 0
- Confirmed-timeout: 0
- Failure: 0
- Type-unsupported: 12

Community Votes

- Harmless: 0
- Malicious: 0

Detailed Engine Results

Engine Name	Detection Category	Detection
No results	No results	No results
